

Dementia Prediction Using Machine Learning

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Abstract—Dementia is a degenerative and chronic condition that may be caused by several diseases that affect memory, thinking, and ability to perform daily activities by destroying nerve cells and damaging the brain over time [1]. This leads to deterioration in cognitive function, beyond the natural consequences of biological aging. Dementia has grown to become a significant health concern for the aged population, and it has become challenging to provide care for patients in the 21st century. The purpose of this work is to develop a framework that allows us to predict dementia using data collected from the OASIS (Open Access Series of Imaging Studies) project which was made available by the Washington University Alzheimer's Disease Research Centre. Multiple techniques were utilized for data imputation, pre-processing, and data transformation to build a suitable database to train our model. Naïve Bayes (NB), Logistic Regression (LR), Decision Tree (DT), K-Nearest Neighbour (KNN), Gradient Boost, XGBoost and LightGBM are among the machine learning approaches used to combine the features by applying them to the full set of features and those selected from Least Absolute Shrinkage and Selection Operator (LASSO) techniques. Upon comparing the accuracy, precision and other metrics on the various classification algorithms' results, Decision trees Algorithm has out-performed all others in our application.

Index Terms—Health Informatics, LASSO, Machine Learning,

I. INTRODUCTION

A. Dementia: Types and Causes

Dementia is recognized as a mental disorder that has a progressive deterioration of cognitive functions that can affect activities like memory, problem-solving, concentration, perception and more [1]. Moreover, dementia is irreversible and progressive [2]. Age, lifestyle, and even brain injuries are key contributors to the development of dementia. However, age is the primary risk factor [1]. Yet Dementia is not a part of the aging process but a cognitive decline that affects your daily life. The primary cause of dementia is the several diseases and injuries that have affected the brain. Millions of people suffer from dementia, making it one of the most significant health challenges in the 21st century. Alzheimer's Disease (AD) and vascular factors are leading causes of dementia, with AD being the most common reason, causing changes in the brain affecting the structural and functional aspects, responsible for 70-80% of cases affecting 50 million people [2]. In addition to this, Dementia took the 7th place as the major cause of death in the older groups globally [1].

Diagnosis is problematic because there is currently no standardized 'dementia test' and also the test is highly specific based on the various classifications of dementia [3]. The various types of Dementia are Frontotemporal Dementia, Lewy

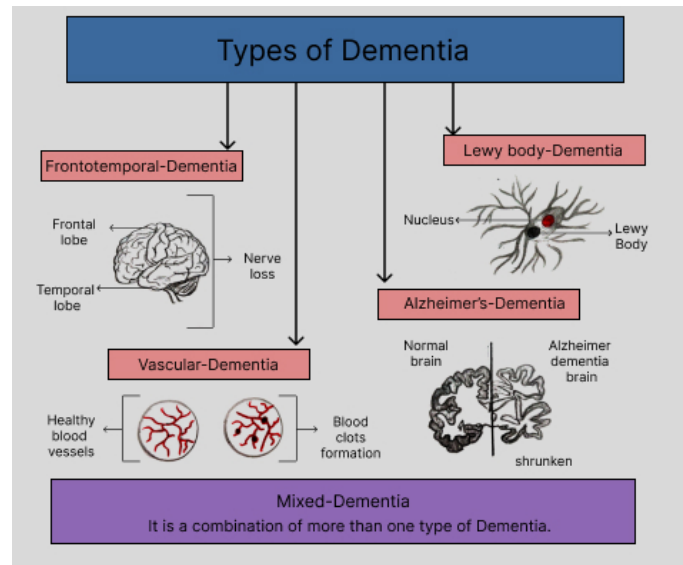


Fig. 1: Types of Dementia

body Dementia, Vascular Dementia, Mixed Dementia, and Alzheimer's Disease. Firstly, Frontotemporal Dementia is a type that is caused by nerve cell loss in the frontal and temporal lobes of the brain [1]. Followed by Lewy body dementia, which is characterized by abnormal deposits of protein alpha-synuclein in the brain. Then Vascular dementia occurs when there are problems with blood flow in the brain. Additionally, Mixed dementia happens when there is more than one kind of dementia [1]. Thus, the chances are that some are the ones who have Mild cognitive impairment (MCI) [3]. The diagnosis of MCI would lead to a higher risk of dementia as the early stage of dementia has already passed and leads to a more serious stage of dementia [3]. Revolutionizing several fields, here too, we aim to use machine learning to identify people who have cognitive complaints, are likely to get dementia or suffer from Mild Cognitive Impairment [2]. The various types of dementia have been outlined in the illustration underneath.

B. Machine Learning in Dementia Prediction

This research aims to develop algorithms that will learn and improve, to be used for predictions. Despite much buzz in the field, proper tests for dementia are yet to exist. It is additionally difficult to distinguish between diagnoses that result in dementia and those that will return to healthy cognitive functioning. Machine Learning makes it possible to identify people with cognitive complaints who are on the verge of dementia or Mild Cognitive Impairment (MCI) [2].

To achieve this goal, Machine Learning (ML), a subset of Artificial Intelligence (AI), can be used as a data analysis technique to computerize the explanatory model structure. Despite the success of both supervised and unsupervised ML algorithms, it was not possible for conventional techniques to analyze the significant volume of information due to time consumption efforts. Thus, becomes even more crucial to find the most suitable technique for our task [4].

ML implementations in healthcare provide efficient methods of using complex information to accurately predict diagnosis. However, in the domain of dementia prediction using ML, there are certain challenges. Due to the prevalence of cognitively normal elderly individuals carrying Alzheimer's disease pathophysiology, it is challenging to accurately identify individuals in the early stages of Alzheimer's. It becomes necessary to identify effective biomarker signatures to gauge the extent of dementia, but accurately predicting the short-term clinical progression has proven difficult [5].

II. RELATED WORK

The use of machine learning, particularly convolutional neural networks (CNN), has become prominent in the classification of Alzheimer's disease (AD) based on medical imaging data, particularly MRI scans. Traditional machine learning models such as logistic regression, support vector machines (SVM), and linear discriminant analysis have been used to predict AD progression. However, these methods are often less effective when faced with complex, imbalanced datasets. CNNs, a deep learning model, have shown promise due to their ability to extract hierarchical features from images without manual feature engineering.

This paper proposes a CNN-based model for multiclass classification of Alzheimer's disease using the Open Access Series of Imaging Studies (OASIS-3) dataset containing 2,168 MRI images. The authors applied 10-fold cross-validation to validate the model and achieved an accuracy of 83.3%, significantly higher than traditional models like logistic regression and SVM. They also compared their work with other state-of-the-art approaches, such as Basheera and Ram (2019), who achieved 90.47% accuracy using a CNN model on T2-weighted MRI images, and Mehmood et al. (2020), who reported a 99.05% accuracy using a Siamese CNN architecture. These studies demonstrate the growing role of CNNs in AD classification, positioning this approach as a leading method for identifying early stages of AD from MRI scans [6].

Quantifying neurodegeneration patterns in Alzheimer's disease (AD) using structural MRI data has gained traction due to advances in machine learning models that can efficiently analyze high-dimensional imaging data. This paper explores various machine learning techniques to analyze and predict AD progression based on structural MRI scans. The research aims to develop models capable of capturing the structural neurodegeneration patterns of the brain, providing more accurate diagnostic tools for clinicians.

Key machine-learning techniques explored in this study include decision trees, random forests, and deep learning models, all of which have been applied to predict AD progression.

These models are compared to determine the most effective approach for quantifying neurodegenerative changes in brain structures associated with AD. Using machine learning, the authors improve the understanding of how different brain regions degenerate over time, contributing to early diagnosis and monitoring disease progression.

The study builds upon existing literature, where machine learning models have demonstrated effectiveness in detecting early AD, using both neuroimaging data and traditional biomarkers. The findings highlight the capability of machine learning models to provide a nuanced understanding of neurodegeneration, enabling more personalized treatment strategies. This study is an important reference for those working on the intersection of computational modeling and neurodegenerative diseases [7].

The Open Access Series of Imaging Studies (OASIS) is a critical resource for neuroimaging research, particularly for Alzheimer's disease (AD) and other neurodegenerative conditions. The OASIS data is widely used in studies that seek to understand the anatomical and functional changes in the brain associated with aging and dementia.

Previous studies using the OASIS dataset have focused on whole-brain volume analysis and longitudinal measures of brain atrophy to track the progression of Alzheimer's. Notably, it has been used to analyze normalized whole-brain volume (nWBV) and the rate of atrophy in individuals with very mild to moderate AD. The dataset supports machine learning research by providing high-contrast MRI data that is ideal for automated analysis, including techniques such as deep learning. Studies like those by Buckner et al. (2004) and Fotenos et al. (2005) have shown that brain atrophy rates in Alzheimer's patients can be accurately measured using OASIS data, establishing it as a foundational tool for computational neuroimaging [8].

III. METHODOLOGY

A. Dataset: OASIS-2

The Open Access Series of Imaging Studies (OASIS) provided the dataset which has been utilized in this study. The objective of the paper is to make brain MRI data sets publicly accessible for further studies and discoveries. The dataset, which we used for our work, comprises longitudinal MRI data belonging to demented and nondemented individuals. 150 subjects between the ages of 60 and 90 who underwent two or more scans make up the longitudinal MRI dataset. Furthermore, there are both right-handed male and female in this collection. Additionally, this collection was divided into 72 subjects who had been categorized as "nondemented," 64 subjects who were classified as "demented," and 14 subjects who were considered as the "Converted Category" for the purpose of first classifying them as "nondemented" and then "demented." [7] As we need to perform binary classification, this discrepancy of three separate labels will be handled in our data pre-processing as discussed in a later section.

B. Discussing Our Approach

Our primary aim in this study is to compare and determine the best classifier approach in order to distinguish 'demented'

from 'nondemented' individuals. To meet this end, we have devised an architecture (shown below), inspired by Dhakal et al. in their paper [2].

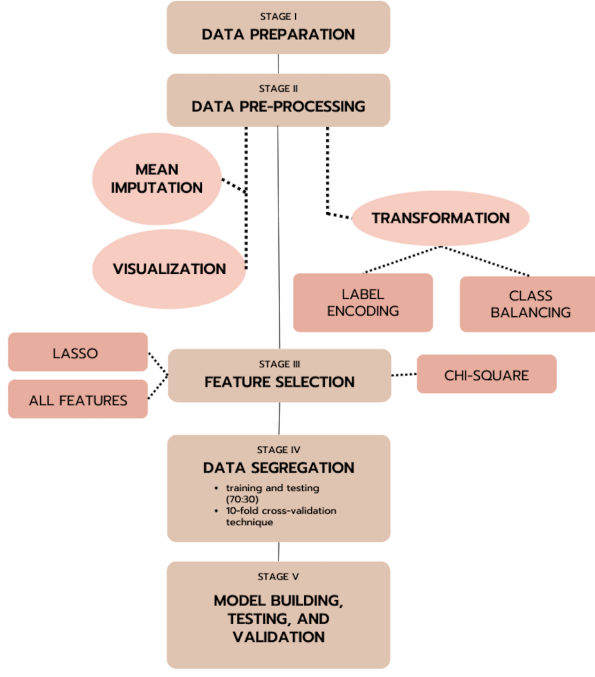


Fig. 2: Proposed model

The proposed model involves pre-processing, feature selection, data segregation, model building, and testing/validations. To improve model performance and generalization, these techniques include mean imputation, transformation, visualization, label encoding, class balancing, Lasso, Chi-Square, 70:30 split, and 10-fold cross-validation.

C. Data Preprocessing: Handling of the missing values

Missing values, noise, and distortions tend to occur in raw datasets that are gathered through a variety of sources, such as the internet, studies, or surveys. Several techniques, including Mean Imputation, Visualization, and Transformation, might be used to solve and view these distorted values.

1) *Mean Imputation*: The technique of substituting missing values with appropriate quantities is known as imputation. Precise data imputation must be accomplished in order to preserve the relationships within the data [9]. Therefore, in this case, we compute the mean of the attributes that contain null data independently, and then we replace the mean value for the null value. Thus, the process of updating null missing values in the data with the mean of that specific characteristic is known as Mean Imputation.

2) *Visualization*: Diagramming the data into a visual representation, such as a scatter or line graphs, heatmaps, and more, is known as visualization. We can more accurately identify and compare the missing null values with the aid of the visualization approach, and we may adjust as required.

3) *Transformation*: Transformation is another type of technique to handle the missing values where we transform the discrepancies which has been occurred in the data. For this, there are two types of transformation ways such as Label encoding and Class Balancing. Label Encoding is a transformation technique where we convert variable categorical columns to numerical ones. In our study for the Dementia Prediction, we have merged the Demented and the Converted categories into one single category. Moreover, in Class Balancing helps us in the removal process of the biases in the data with the help of undersampling and oversampling. Here, Undersampling is the process where the tuples in the data are decreased to reduce the bias nature whereas Oversampling is the process where the tuples are increased in the data to reduce the biasing nature. In our study, we have done undersampling in the Demented category as the Demented category's ratio was more than the other categories. Therefore, to remove the biases towards the Demented category, the process of undersampling was done onto it.

D. Feature Selection

Feature selection is the method of choosing the features to be used and removing input features that are not important for classification [10] [11]. There are certain kinds of feature selection algorithms such as Least Absolute Shrinkage and Selection Operator (LASSO), All Features and Chi-Square which are sequentially described below.

1) *Least Absolute Shrinkage and Selection Operator (LASSO)*: LASSO is a shrinkage method that performs both variable selection and regularization [12]. This method helps prevent the over fitting of the predictive model to the data.

2) *All Features*: All Features method is certainly not a method but a way where we use all the features from the dataset provided for the prediction of the model.

3) *Chi-Square*: The Chi square test is used in statistics to measure the lack of correlation between the features [11] The formula of the Chi-Square is: $\chi^2 = \sum \frac{(O - E)^2}{E}$ where O is Observed and E is Expected. The more the difference of the Observed and the Expected in the formula of the Chi-Square, The higher is the value of the Chi-Square [13].

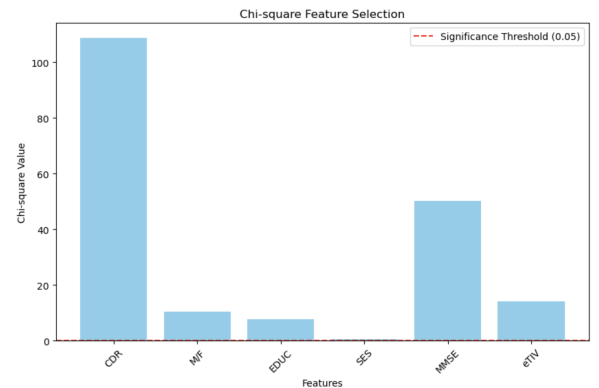


Fig. 3: Features selected using Chi-Square technique

E. Data Segregation

A further phase is data segregation, where the feature-selected and pre-processed data is segregated. Overfitting and underfitting problems are prevented by using this method [2]. Here, we have divided the data in a 70:30 ratio, with 70 embarking to the train and 30 going to the data testing. The problem of generalization is resolved by this data separation. In addition to data splitting, we employed K-fold cross-validation in our investigation, using $k=10$ in the training set. By splitting the training model into ten samples and cross-training and testing them, the 10-fold cross-validation procedure aids in improved validation. Consequently, it offers superior validation [2].

F. Model Building

All of the aforementioned stages applied to this stage of model building. Training an algorithm to predict the label target variable from the characteristics independent variables is known as model building. In this procedure, the algorithm receives the training data, which it utilizes to learn and create the model. Additionally, the train data was fed into a classifier in order for it to provide a valid output. This method is referred to as model fitting as the outcome is now utilized to forecast the test data [1]. The target variables in our study include two values: demented and non-demented. To predict dementia, we employed a range of machine learning techniques to our dataset, including Naïve Bayes, Logistic Regression, K nearest neighbour, Decision Trees, Gradient Boost Classifier, Extreme Gradient Boosting and Light Gradient Boosting.

IV. ANALYSIS OF THE RESULT

A. Validation score for all features

Table I to VII shows various algorithms and its accuracy with all features. If we observe it from the highest to lowest accuracy then Decision tree has the highest accuracy of 1.00, XGBoost Classifier has the accuracy of 0.93, followed by Naïve Bayes with accuracy of 0.91, then Light GB Classifier with an accuracy of 0.91, later Logistic Regression with 0.90 and with the least accuracy of 0.59 is of the KNN Classifier.

TABLE I: Naïve Bayes

Class	Precision	Recall	F1-Score
Demented (0)	0.96	0.87	0.91
Non-Demented (1)	0.86	0.96	0.91
Accuracy	0.91		

TABLE II: Logistic Regression

Class	Precision	Recall	F1-Score
Demented (0)	0.98	0.83	0.90
Non-Demented (1)	0.84	0.98	0.90
Accuracy	0.90		

TABLE III: K-NN

Class	Precision	Recall	F1-Score
Demented (0)	0.64	0.65	0.64
Non-Demented (1)	0.52	0.50	0.51
Accuracy	0.59		

TABLE IV: Decision Tree

Class	Precision	Recall	F1-Score
Demented (0)	1.00	1.00	1.00
Non-Demented (1)	1.00	1.00	1.00
Accuracy	1.00		

TABLE V: Gradient Boosting Classifier

Class	Precision	Recall	F1-Score
Demented (0)	0.98	0.88	0.93
Non-Demented (1)	0.88	0.98	0.93
Accuracy	0.92		

TABLE VI: XGBoost Classifier

Class	Precision	Recall	F1-Score
Demented (0)	0.95	0.92	0.93
Non-Demented (1)	0.91	0.94	0.92
Accuracy	0.93		

TABLE VII: LightGB Classifier

Class	Precision	Recall	F1-Score
Demented (0)	0.97	0.86	0.91
Non-Demented (1)	0.84	0.97	0.90
Accuracy	0.91		

B. Validation score for Chi-2 selected features

Table VIII to XIII shows various accuracies of the same algorithms but with Chi-Square features. If we observe them according to the most to least of the accuracies, we see Decision tree with the highest accuracy of 1.00, followed by Gradient Boosting Classifier with 0.93 and XGBoost Classifier with 0.93 accuracy, then Logistic regression with 0.92 accuracy, Naïve Bayes with the accuracy of 0.92 and the least accuracy of 0.79 which is of the KNN.

TABLE VIII: Naïve Bayes

Class	Precision	Recall	F1-Score
Demented (0)	0.95	0.87	0.90
Non-Demented (1)	0.86	0.94	0.90
Accuracy	0.92		

TABLE IX: Logistic Regression

Class	Precision	Recall	F1-Score
Demented (0)	0.98	0.87	0.92
Non-Demented (1)	0.86	0.98	0.92
Accuracy	0.92		

TABLE X: K-NN

Class	Precision	Recall	F1-Score
Demented (0)	0.77	0.88	0.82
Non-Demented (1)	0.84	0.69	0.76
Accuracy	0.79		

TABLE XI: Decision Tree

Class	Precision	Recall	F1-Score
Demented (0)	1.00	1.00	1.00
Non-Demented (1)	1.00	1.00	1.00
Accuracy	1.00		

TABLE XII: Gradient Boosting Classifier

Class	Precision	Recall	F1-Score
Demented (0)	0.98	0.88	0.93
Non-Demented (1)	0.88	0.98	0.93
Accuracy	0.93		

TABLE XIII: XGBoost Classifier

Class	Precision	Recall	F1-Score
Demented (0)	0.95	0.92	0.93
Non-Demented (1)	0.91	0.94	0.92
Accuracy	0.93		

TABLE XIV: LightGB Classifier

Class	Precision	Recall	F1-Score
Demented (0)	0.97	0.86	0.91
Non-Demented (1)	0.84	0.97	0.90
Accuracy	0.91		

C. Validation score for LASSO selected features

Table XIV to XXI also shows the accuracies of the above discussed algorithms with LASSO features. If we observe the model from the most to least accuracy values then we get Decision tree with the highest accuracy of 0.98, followed by Gradient Boosting with accuracy 0.93, then XG Boost classifier of 0.92 accuracy, Naïve Bayes and Logistic Regression with 0.92 accuracy, then LightGB Classifier with 0.91 accuracy and the least accuracy is of KNN with 0.66.

TABLE XV: Naïve Bayes

Class	Precision	Recall	F1-Score
Demented (0)	0.98	0.87	0.92
Non-Demented (1)	0.86	0.98	0.92
Accuracy	0.92		

TABLE XVI: Logistic Regression

Class	Precision	Recall	F1-Score
Demented (0)	0.98	0.87	0.92
Non-Demented (1)	0.86	0.98	0.92
Accuracy	0.92		

TABLE XVII: K-NN

Class	Precision	Recall	F1-Score
Demented (0)	0.61	0.72	0.66
Non-Demented (1)	0.70	0.58	0.63
Accuracy	0.66		

TABLE XVIII: Decision Tree

Class	Precision	Recall	F1-Score
Demented (0)	0.99	0.97	0.98
Non-Demented (1)	0.97	0.99	0.98
Accuracy	0.98		

TABLE XIX: Gradient Boosting Classifier

Class	Precision	Recall	F1-Score
Demented (0)	0.98	0.88	0.93
Non-Demented (1)	0.88	0.98	0.93
Accuracy	0.93		

TABLE XX: XGBoost Classifier

Class	Precision	Recall	F1-Score
Demented (0)	0.95	0.90	0.92
Non-Demented (1)	0.89	0.94	0.92
Accuracy	0.92		

TABLE XXI: LightGB Classifier

Class	Precision	Recall	F1-Score
Demented (0)	0.97	0.86	0.91
Non-Demented (1)	0.84	0.97	0.90
Accuracy	0.91		

V. RESULT INTERPRETATION

A. Challenges and their approaches

The KNN machine learning algorithm's inadequate precision and the decision tree's overfitting problem were two of the difficulties encountered during the aforementioned procedure

and the outcomes analysis. The accuracy of the KNN machine learning method is very low—below 0.50—when the value of k is randomly chosen to compute the accuracy. Therefore, we tested with several choices of k before using the value $k=10$ to get over the hurdle. Despite this, it remains a challenge because its accuracy ranges from 0.60 to 0.80. An further challenge is the decision tree's overfitting, as the algorithm achieves 1.00 accuracy in every model-building procedure. We therefore employed a few more algorithms that make use of the decision tree notion in order to solve this difficulty. These algorithms—which include Gradient Boosting, Light GBM, and others—get around the overfitting problem and offer accuracy between 0.90 and 1.00.

B. Inferences:

The Decision Tree classifier consistently achieves a perfect accuracy of 1.00 in both the full features and Chi-squared selected features, indicating it perfectly classifies all instances in the validation set. The Gradient Boosting Classifier also performs very well, achieving accuracies of 0.92 to 0.93 across different feature selections. Naïve Bayes and Logistic Regression yield similar accuracies of around 0.90 to 0.92. At the same time, K-NN shows the weakest performance with an accuracy ranging from 0.59 to 0.66, suggesting it struggles more than the other models. Precision indicates the accuracy of positive predictions. The Naïve Bayes and Logistic Regression models show high precision for the Demented class (0) but have a moderate decrease for Non-Demented (1) in some cases. Similarly, recall indicates how well the models capture all actual positive instances. The Decision Tree consistently exhibits perfect recall (1.00) for both classes, showing it does not miss any positive cases. K-NN, however, has the lowest recall for Non-Demented (1), indicating it struggles to correctly identify these instances. The Gradient Boosting Classifier maintains a balanced performance in both precision and recall across different feature selections. The F1-Score balances precision and recall, and it is particularly high (around 0.92 - 0.93) for the Gradient Boosting Classifier, Logistic Regression, and Naïve Bayes. The Decision Tree again stands out with a perfect F1-Score. K-NN has a lower F1-Score, indicating its performance is less reliable than the other models, especially in terms of making balanced predictions.

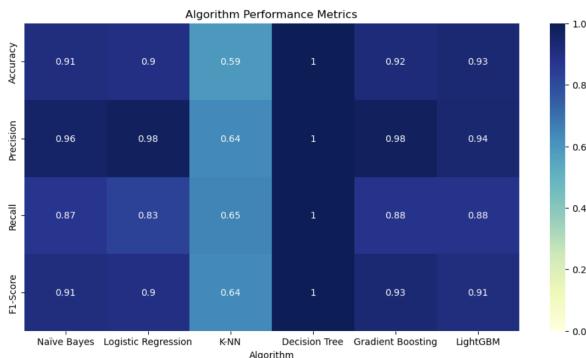


Fig. 4: Heatmap of performances

The aforementioned Figure:4 shows the heatmap about the performances of different machine learning algorithms across various evaluation metrics. The x axis represents the algorithms such as Naive Bayes, KNN, Decision trees along more and the y axis represents the Performance metrics which is the accuracy, precision, recall and F1score. Along with this, the colours of the heatmap represents its performance level- Dark Blue represents excellent performance, Moderate blue represents moderate or average performance and Light blue represents poor performance pf various algorithms. Therefore, from the heatmap we can observe that the Decision Tree has the darkest shade of blue thus having the highest accuracy of 1.00, followed by LightGBM having a shade lighter than Decision Trees' thus having the second highest accuracy of 0.93, then Gradient Boosting with 0.92 and then Naive Bayes and Logistic Regression with 0.91 and 0.9 respectively and the lightest shade of blue is of KNN thus having lowest accuracy of 0.59.

Impact of feature selection: Chi-Squared Selected Features: Performance metrics slightly improve for most algorithms when using Chi-squared selected features, indicating that the selected features help the models to generalize better. The Decision Tree maintains its perfect classification even with selected features, demonstrating its robustness against overfitting in this context.

C. Conclusion

Best Performers: The Decision Tree and Gradient Boosting Classifier are the most reliable algorithms for this classification task, showing excellent accuracy, precision, recall, and F1-Score across different feature sets. **Weak Performer:** The K-NN algorithm struggles in terms of accuracy and recall, particularly in identifying the Non-Demented class. Naïve Bayes and Logistic Regression are solid choices as well, with reasonable performance that is likely suitable for applications requiring interpretability and speed.

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